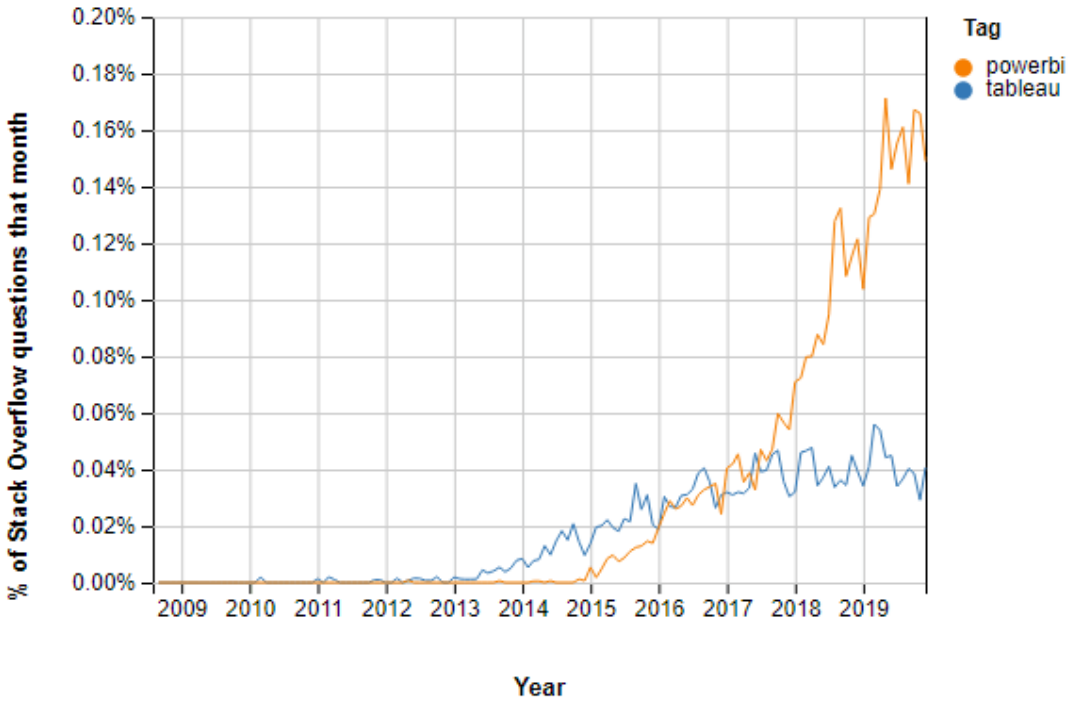
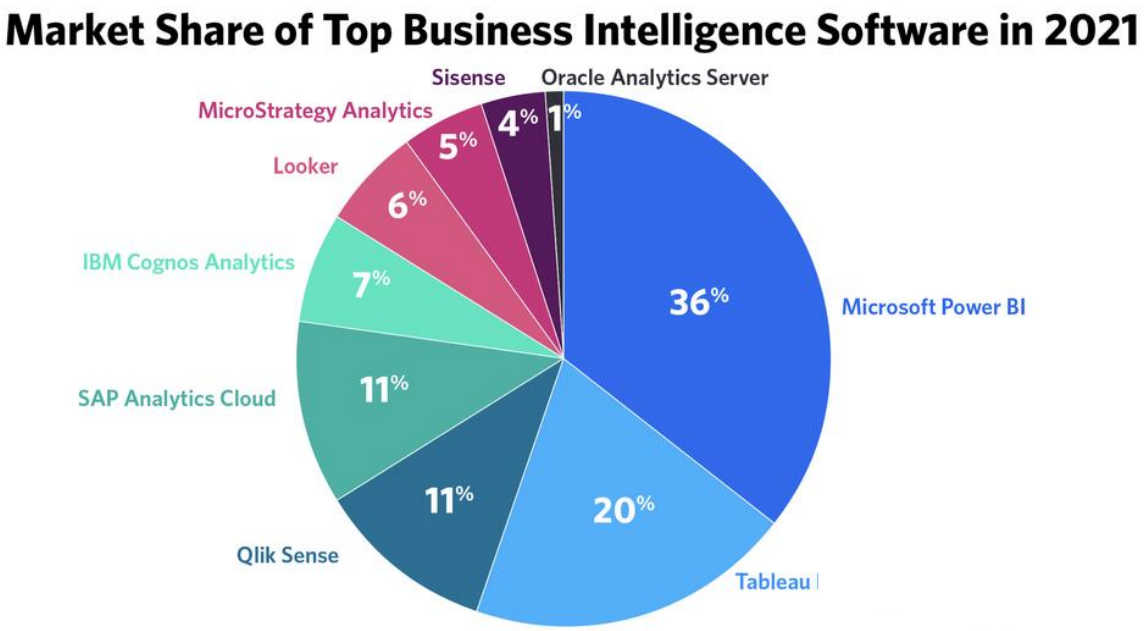
Power BI vs Tableau





Tableau

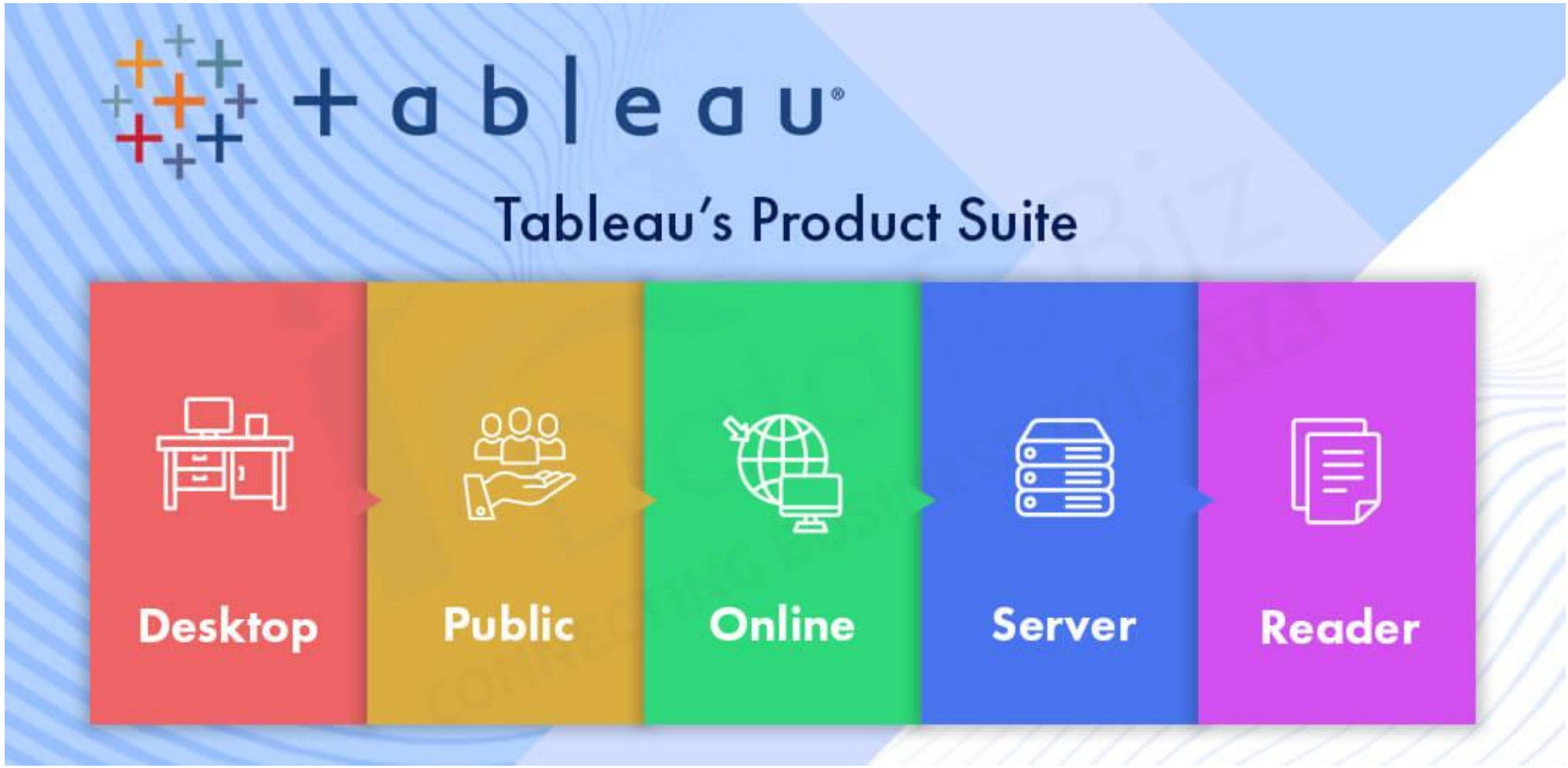


tableau
Tableau's Product Suite
Desktop
Public
Online
Server
Reader

Tableau Products

Tableau Product Suite

	Desktop		Reader	Server	Public	Online
	Personal	Professional				
Details	- Local client for building dashboards - Limited data sources, no ability to connect to Tableau Server	- Local client for building dashboards - Full enterprise capabilities	- Local client to view and interact with local files - Unable to modify workbooks or connect to server	- Privately managed Tableau Server (may be on premise or service hosted) - Users may directly interact with dashboards via browser	- Essentially a massive, public non-commercial Tableau server - All data published is public - Free client available to create dashboards	- Private version of Tableau Public eliminates need for infrastructure - Live connections currently only possible with Google BigQuery and Amazon Redshift
OS					N/A	N/A
License	$999	$1,999	Free	Named User or Core Licensing	Free	$500/user per year

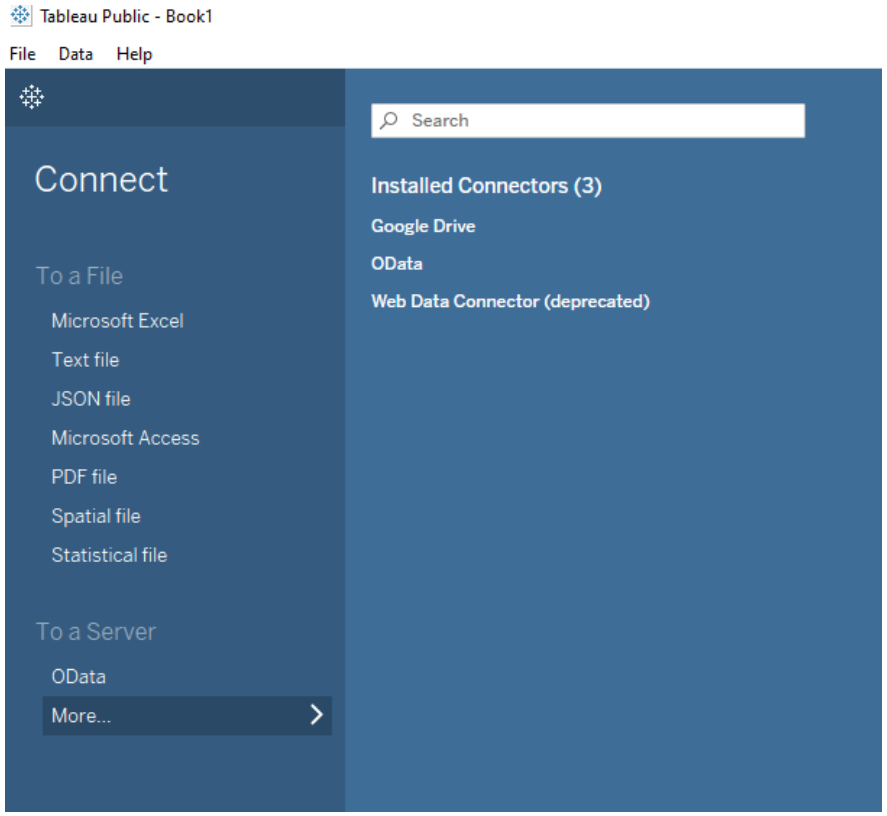
Other Tableau Products

- Tableau GPT
  - GPT
- Tableau Prep
  - ETL
- Tableau CRM Analytics
  - CRM
- Tableau Data Management
  - Data warehousing
- Tableau Advanced Management
  - Enterprise account management for Tableau products

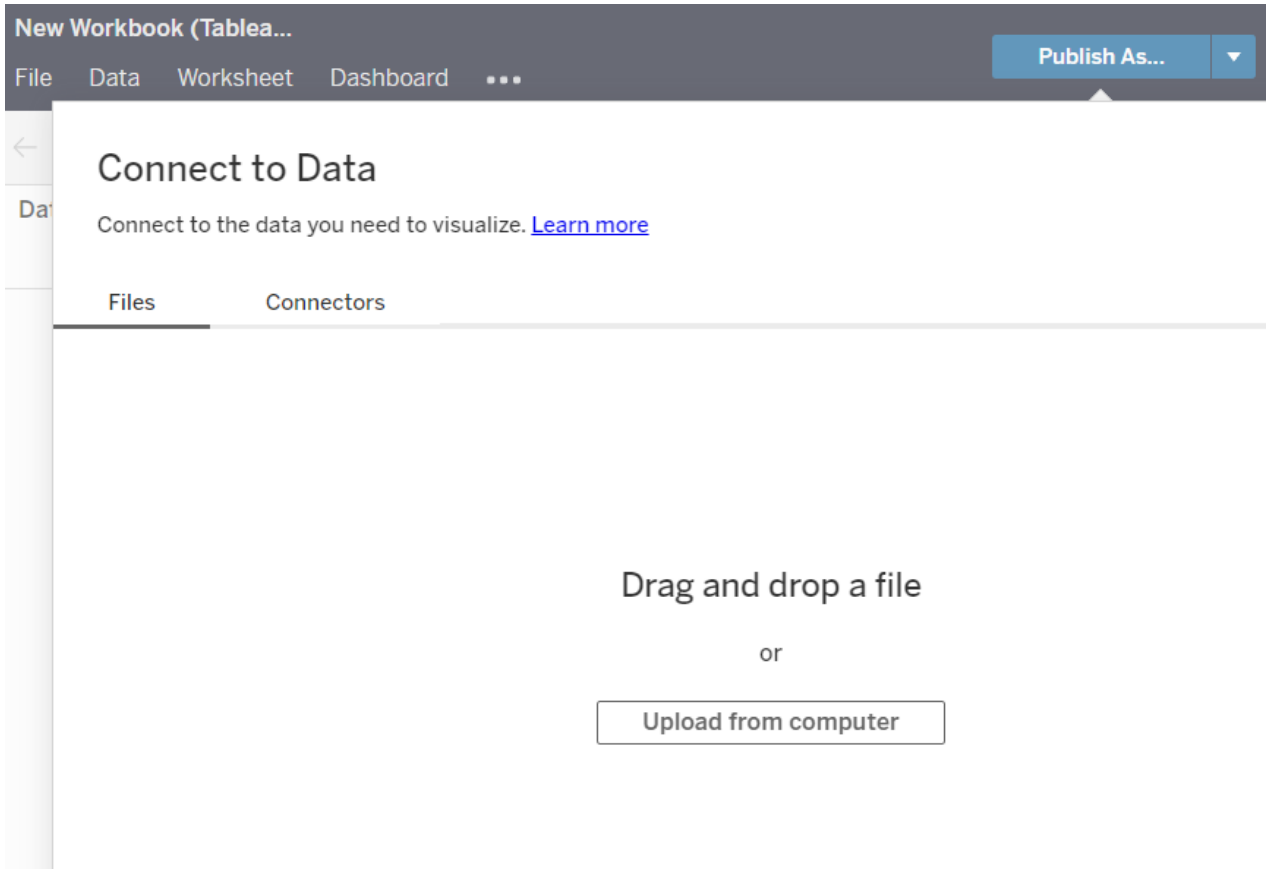
Tableau User Roles

- Creator
  - Data scientists / business intelligence specialists
- Viewer
  - Business users / managers
- Explorer
  - A subset of Creator capability
    - Working on existing data sources and workbooks
  - Cheaper licensing fee than Creator

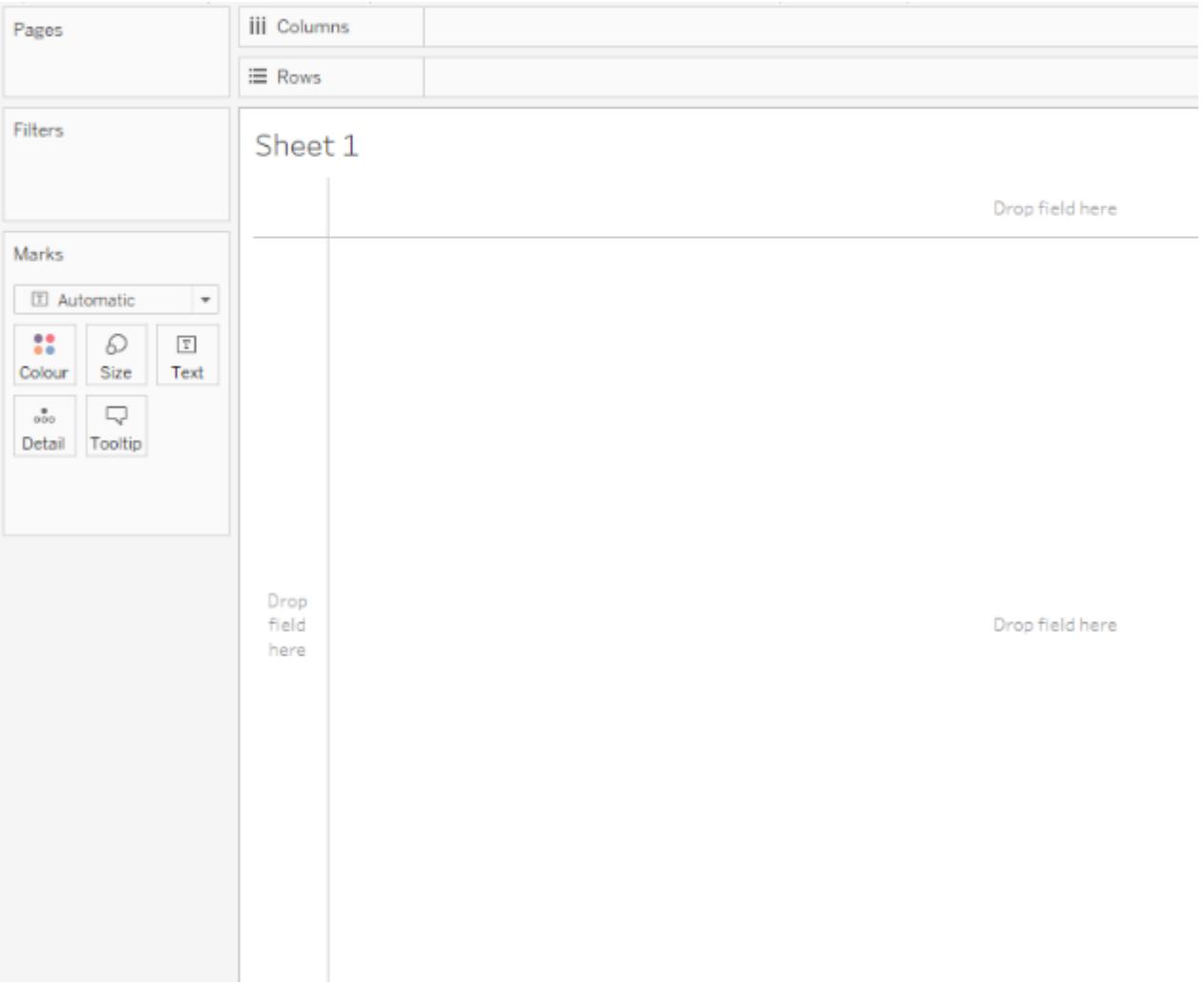
Data Sources on Tableau Public Desktop

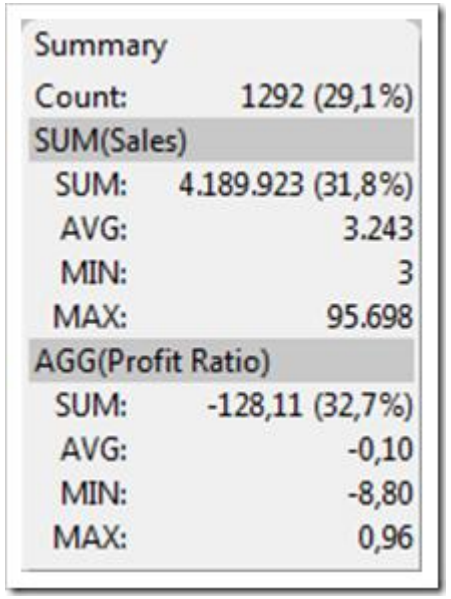


Data Sources on Tableau Public Cloud



Worksheet, Canvas, View, Shelves and Cards

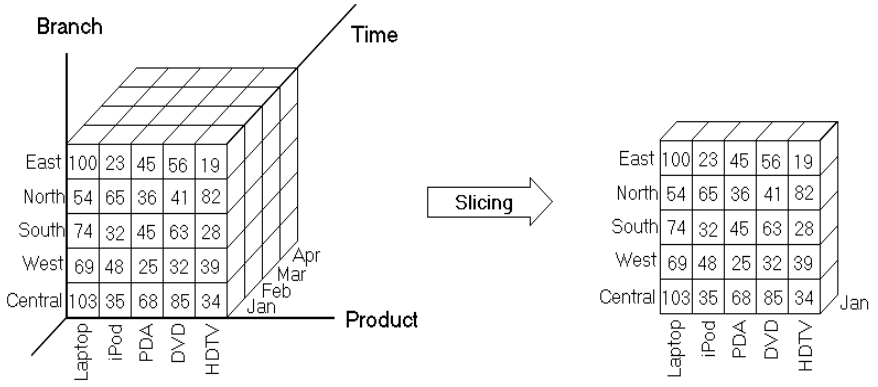




Summary	
Count:	1292 (29,1%)
SUM(Sales)	
SUM:	4.189.923 (31,8%)
AVG:	3.243
MIN:	3
MAX:	95.698
AGG(Profit Ratio)	
SUM:	-128,11 (32,7%)
AVG:	-0,10
MIN:	-8,80
MAX:	0,96

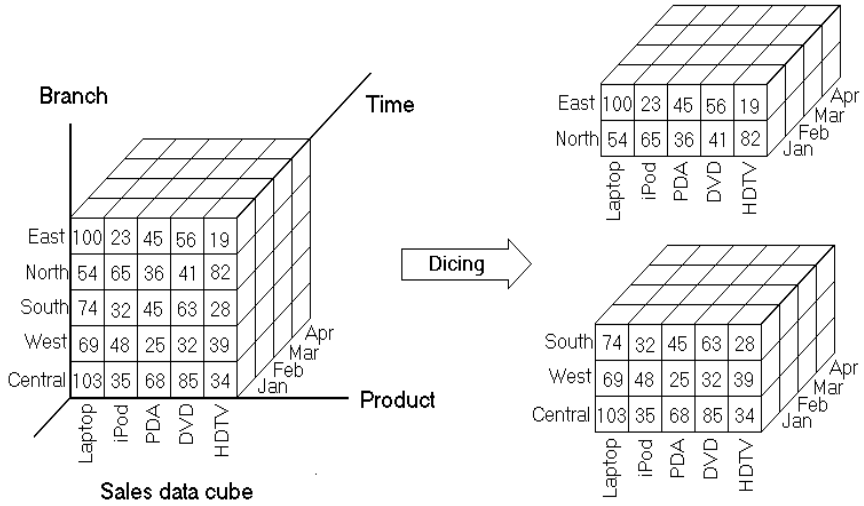
Multi-dimensional Data Model

- **Measures**
  - Numerical data that can be analyzed and compared, such as sales or revenue
- **Dimensions**
  - Dimensions are attributes that describe the measures, such as time, location, or product
- **Hierarchies**
  - A way of organizing dimensions into different levels of detail and granularity
  - For example, a time dimension might be organized into years, quarters, months, weeks, and days
  - Hierarchies provide a way to navigate the data and perform drill-down and roll-up operations
- **Aggregation**
  - Summarize data across different dimensions as well as different levels of detail and granularity
- **Drill-down**
  - Move from a higher-level aggregation of data to a lower level of detail
- **Roll-up**
  - Move from a lower-level detail to a higher-level summary in order to gain insights into the underlying patterns
- **Data cube**
  - A multi-dimensional array corresponding to a multi-dimensional data model
- **Slicing**
  - Reduce the dimensionality of data by fixing the level(s) of one or more dimensions
- **Dicing**
  - Divide a data cube into sub-cubes (called dice) for comparison



Sales data cube

Slice of Jan.



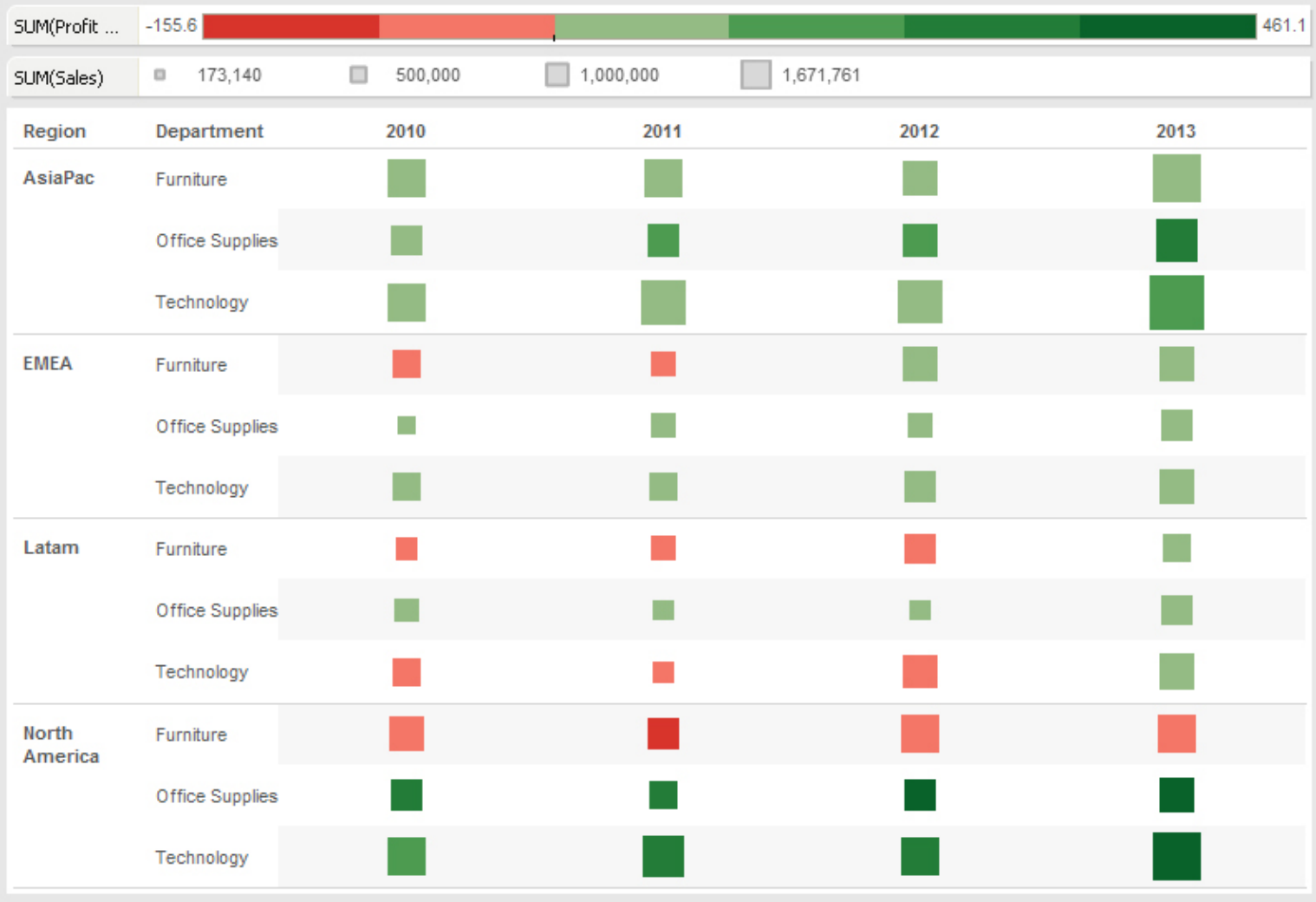
Sales data cube

Two Dices

Tableau Charts

- Tableau vs Microsoft Power BI
  - Microsoft Power BI – choose a visual first
  - Tableau – create a multi-dimensional data model first
- Tableau charts

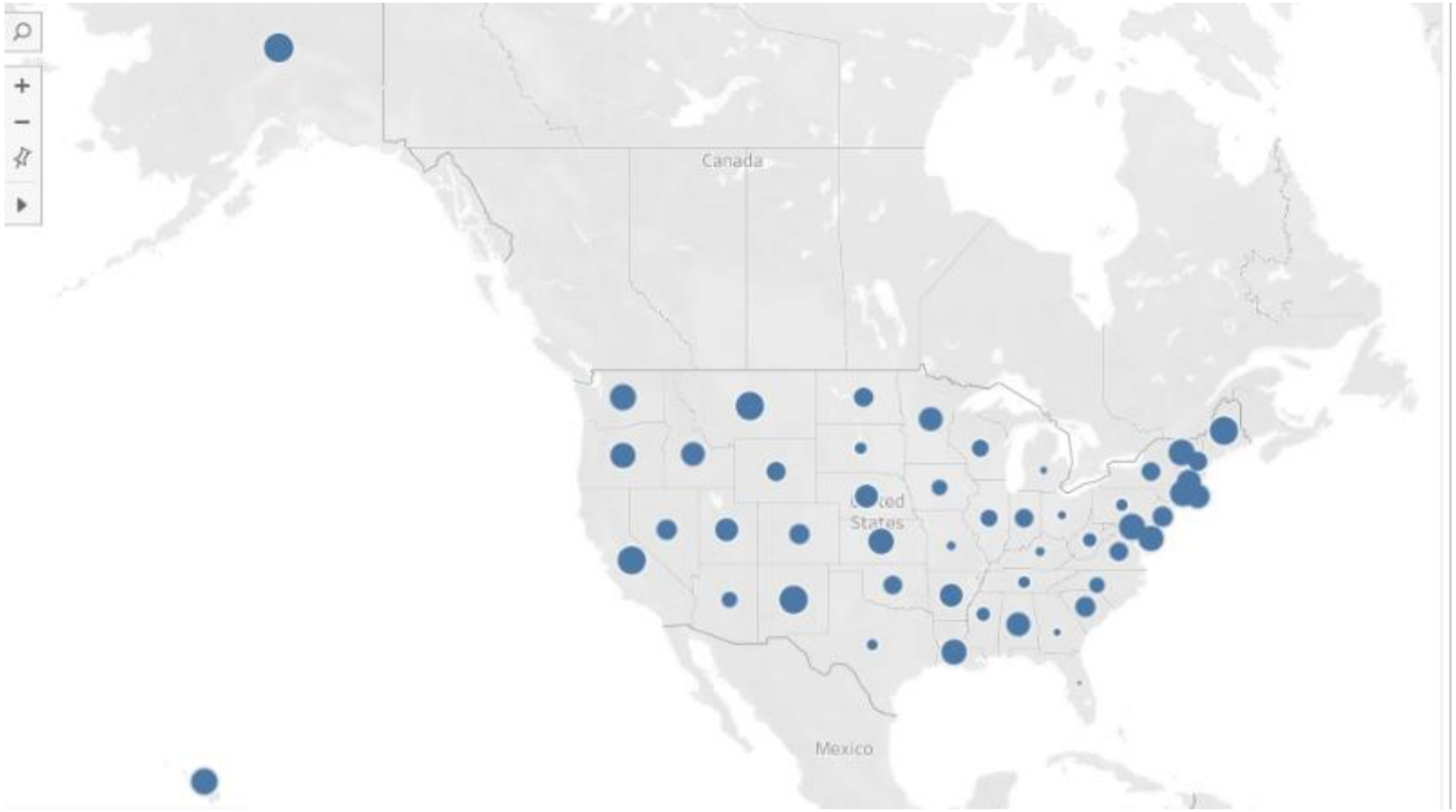
Heat Map


SUM(Profit ...
-155.6
461.1
SUM(Sales)
173,140
500,000
1,000,000
1,671,761
Region
Department
2010
2011
2012
2013
AsiaPac
Furniture
Office Supplies
Technology
EMEA
Furniture
Office Supplies
Technology
Latam
Furniture
Office Supplies
Technology
North America
Furniture
Office Supplies
Technology

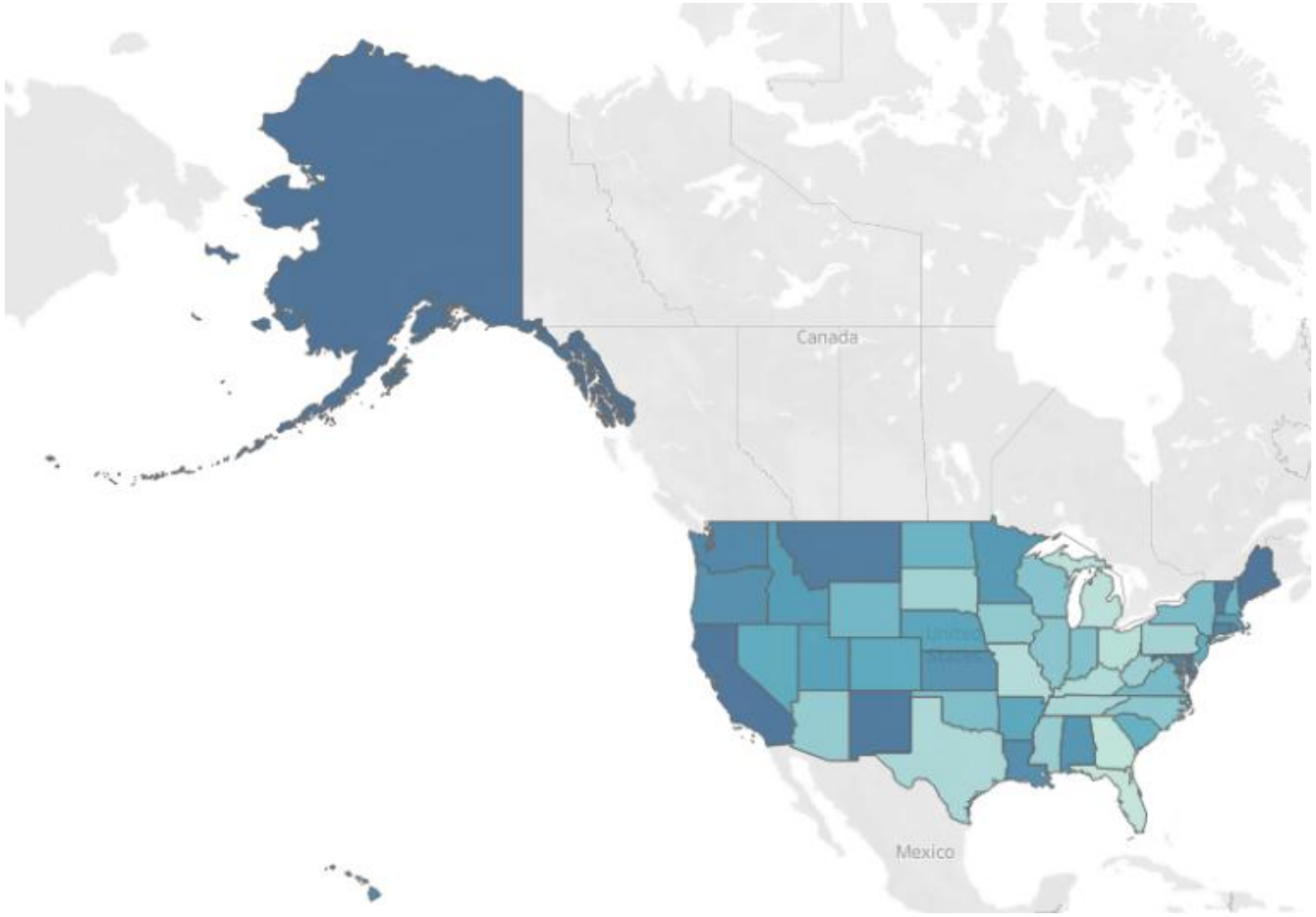
Highlight Table

Sub-Category	2017	2018	2019	2020
Accessories	$6,403	$10,197	$9,664	$15,672
Appliances	$2,459	$2,512	$5,301	$7,865
Art	$1,407	$1,485	$1,414	$2,222
Binders	$4,740	$7,597	$10,216	$7,670
Bookcases	($346)	($2,755)	$212	($584)
Chairs	$6,955	$6,228	$5,763	$7,644
Copiers	$2,913	$9,930	$17,743	$25,032
Envelopes	$1,495	$1,960	$2,067	$1,442
Fasteners	$179	$172	$294	$305
Furnishings	$1,973	$3,052	$3,935	$4,099
Labels	$1,286	$1,323	$1,193	$1,745
Machines	$369	$2,977	$2,907	($2,869)
Paper	$6,371	$6,570	$9,072	$12,041
Phones	$11,808	$10,399	$9,460	$12,849
Storage	$4,166	$3,505	$6,204	$7,403
Supplies	$490	($25)	($699)	($955)
Tables	($3,124)	($3,510)	($2,951)	($8,141)

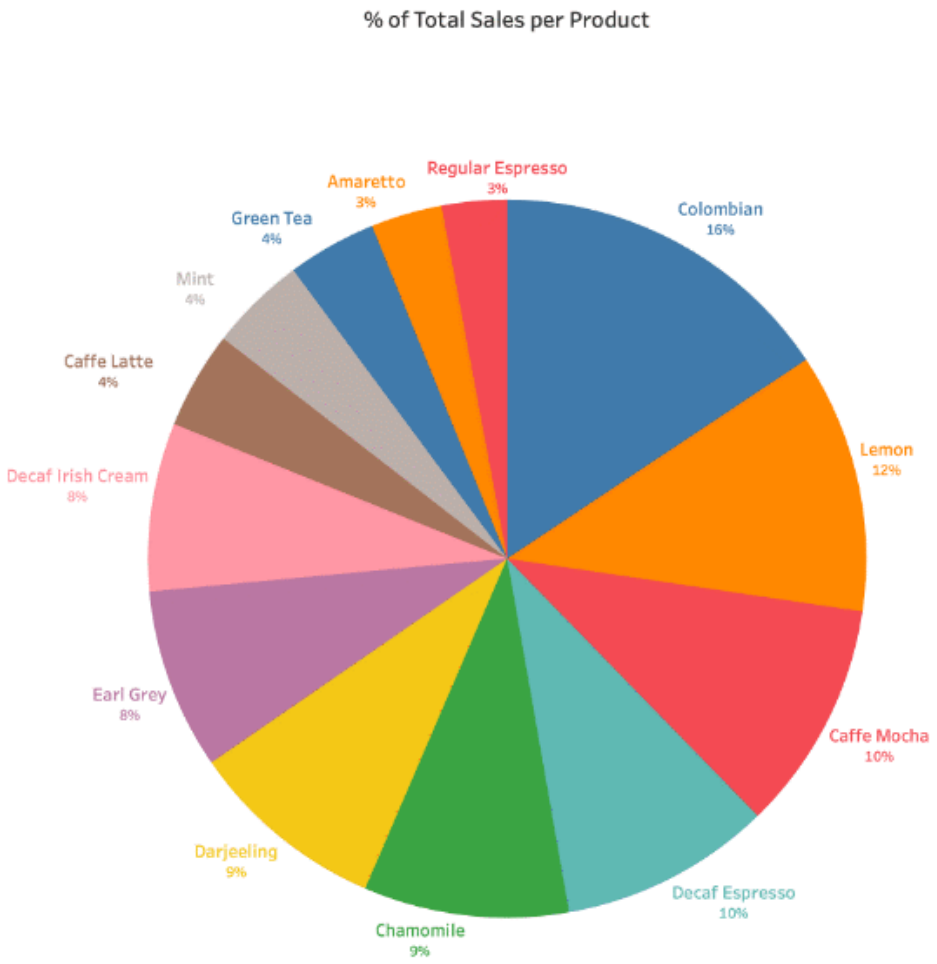
Symbol Map



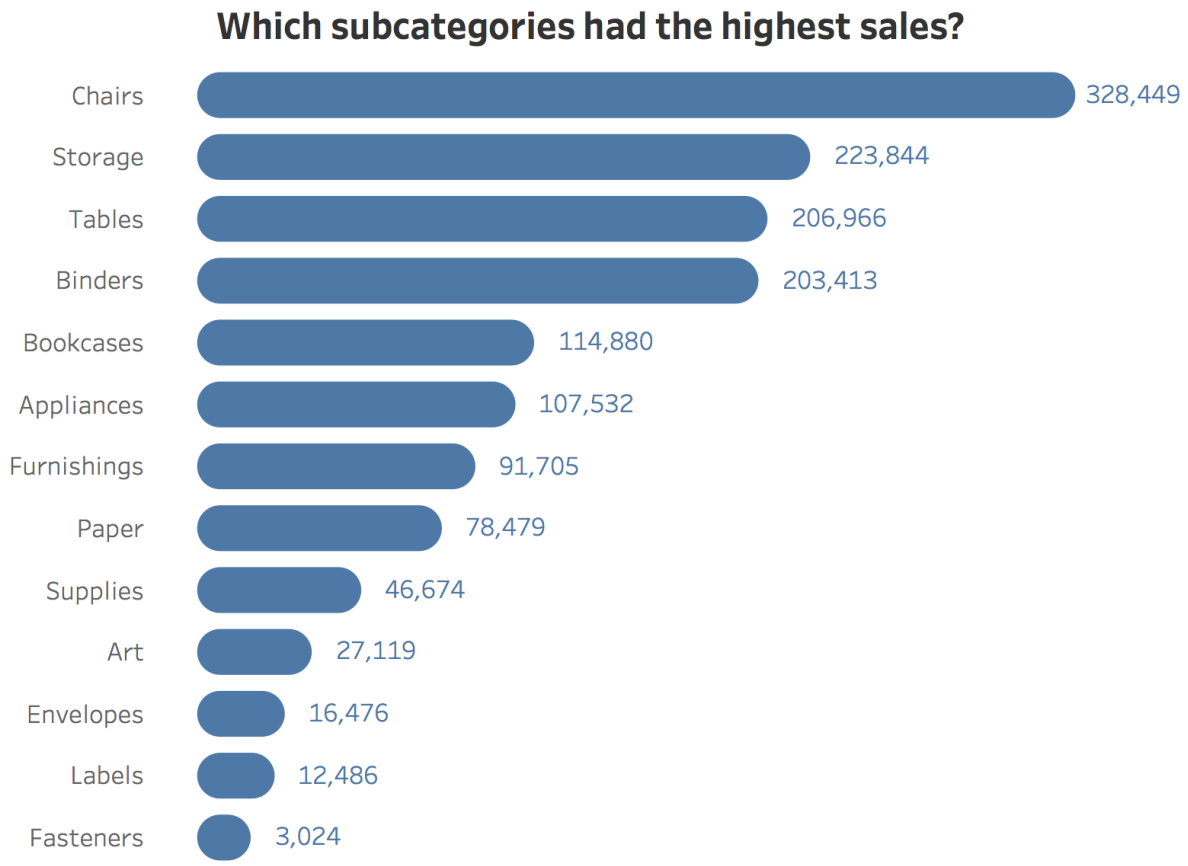
Map



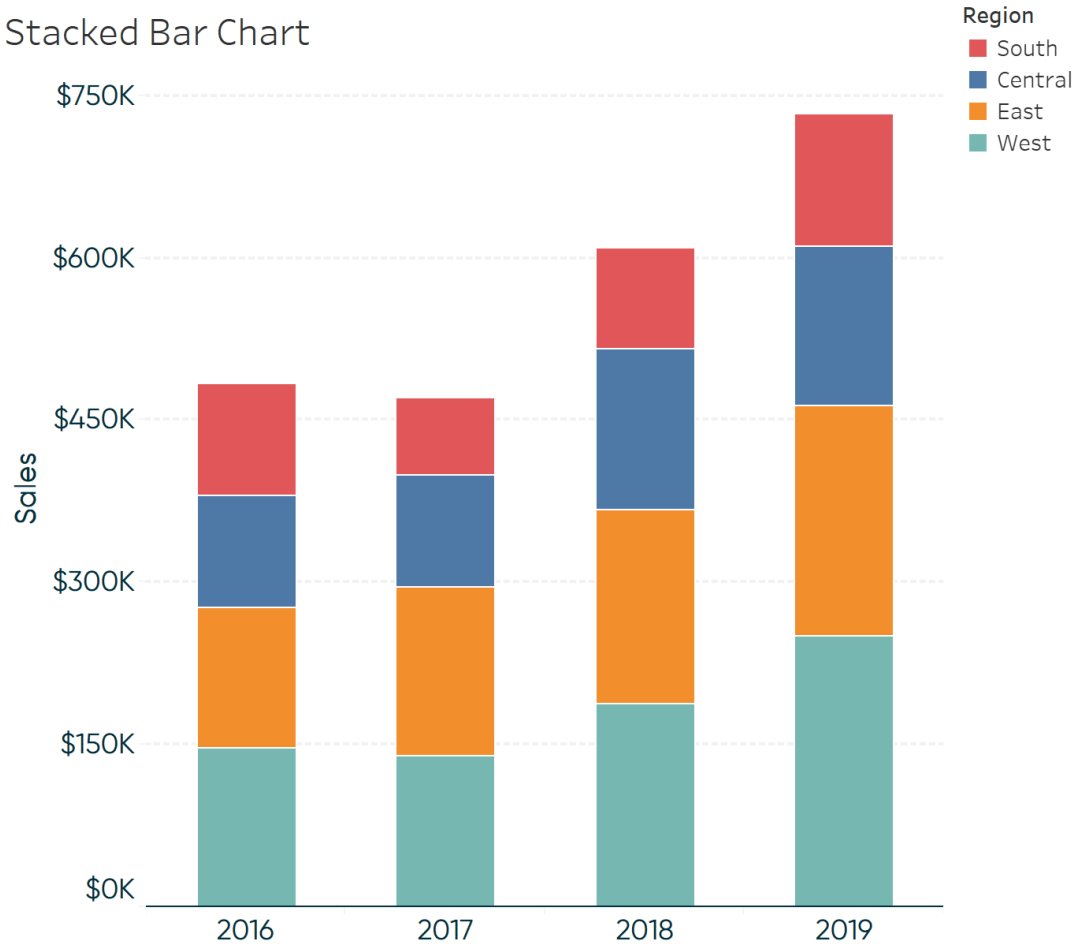
Pie Chart


% of Total Sales per Product
Regular Espresso
3%
Amaretto
3%
Green Tea
4%
Mint
4%
Caffe Latte
4%
Decaf Irish Cream
8%
Earl Grey
8%
Darjeeling
9%
Chamomile
9%
Decaf Espresso
10%
Caffe Mocha
10%
Lemon
12%
Colombian
16%

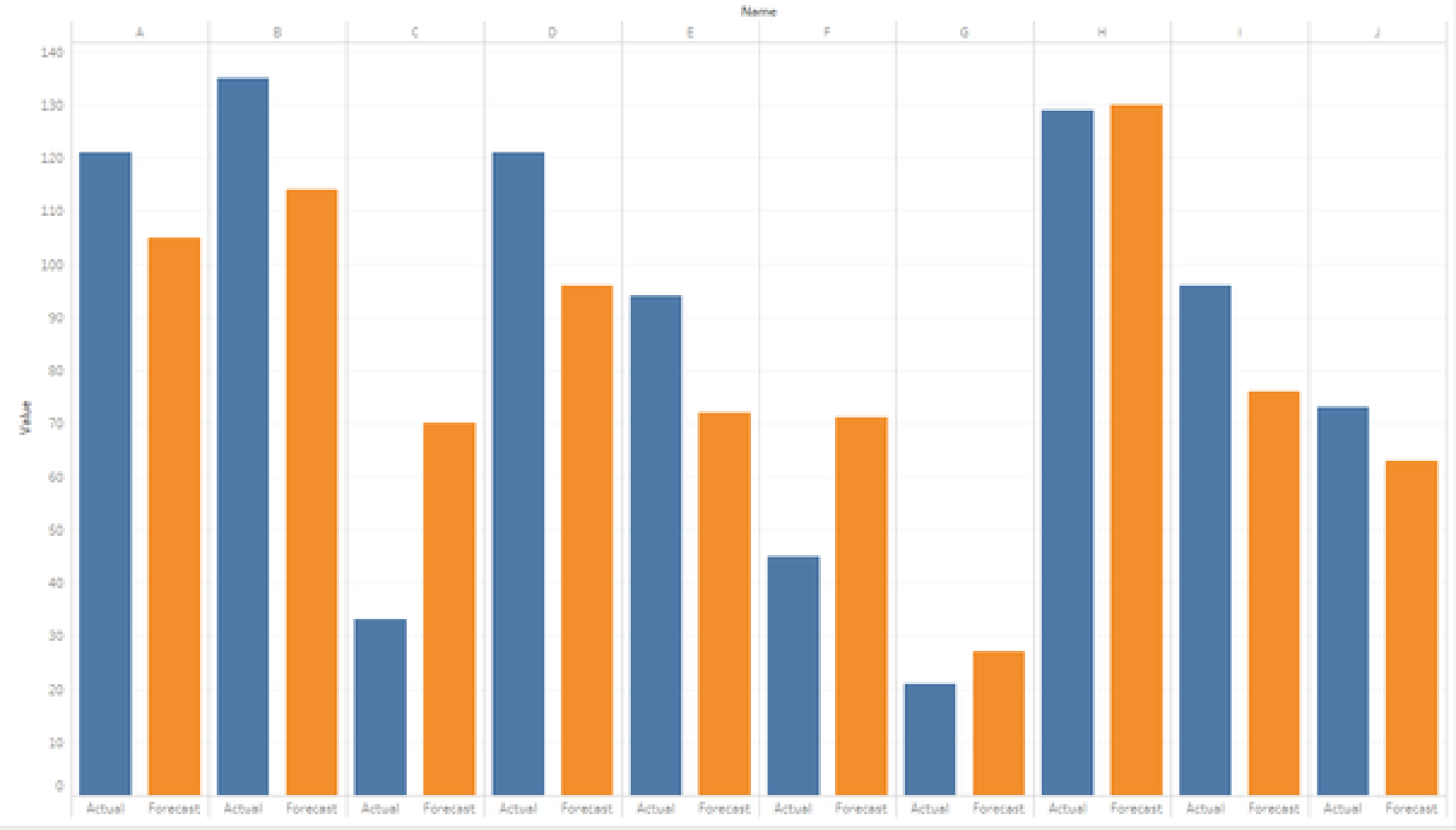
Horizontal Bar Chart



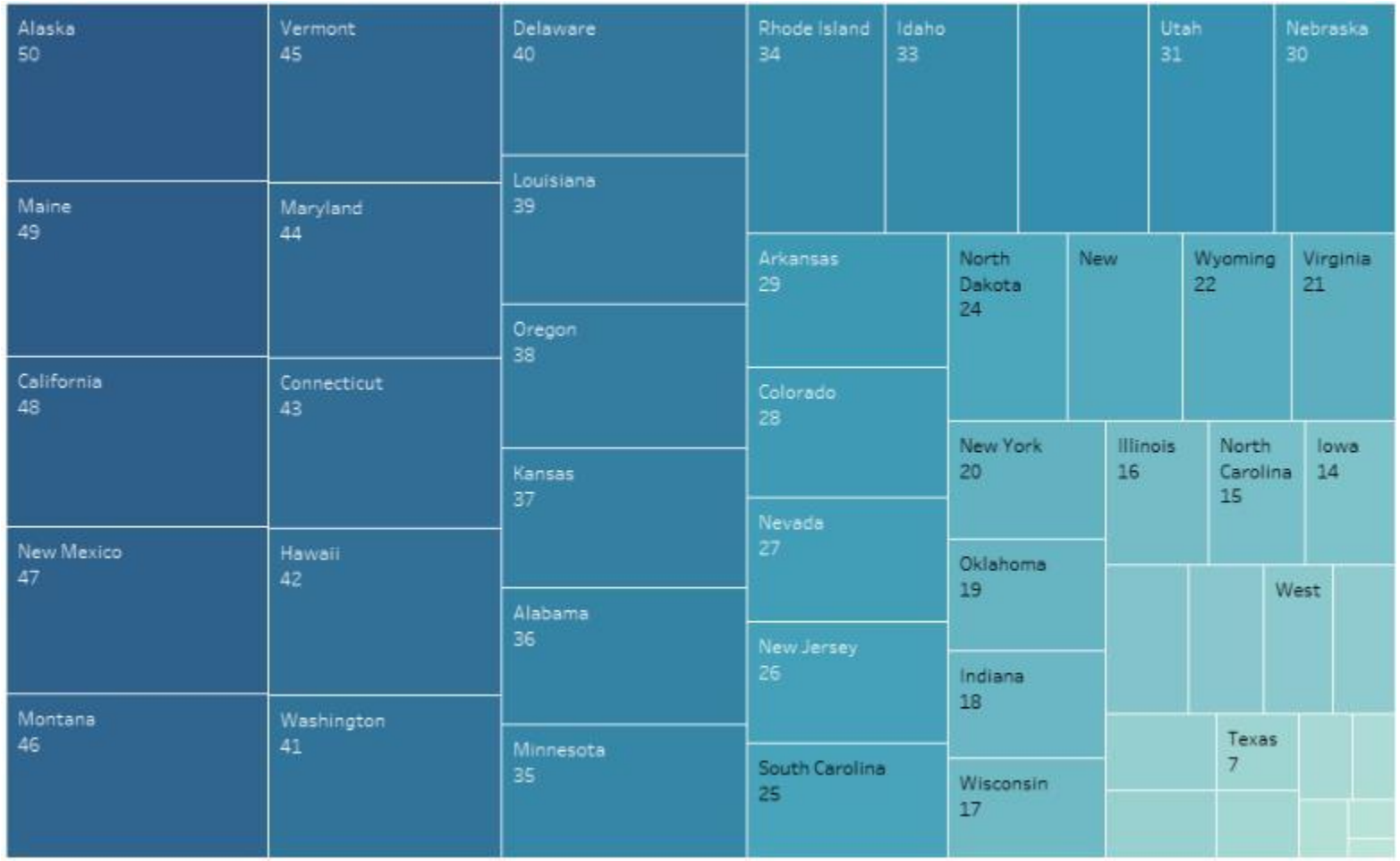
Stacked Bar Chart



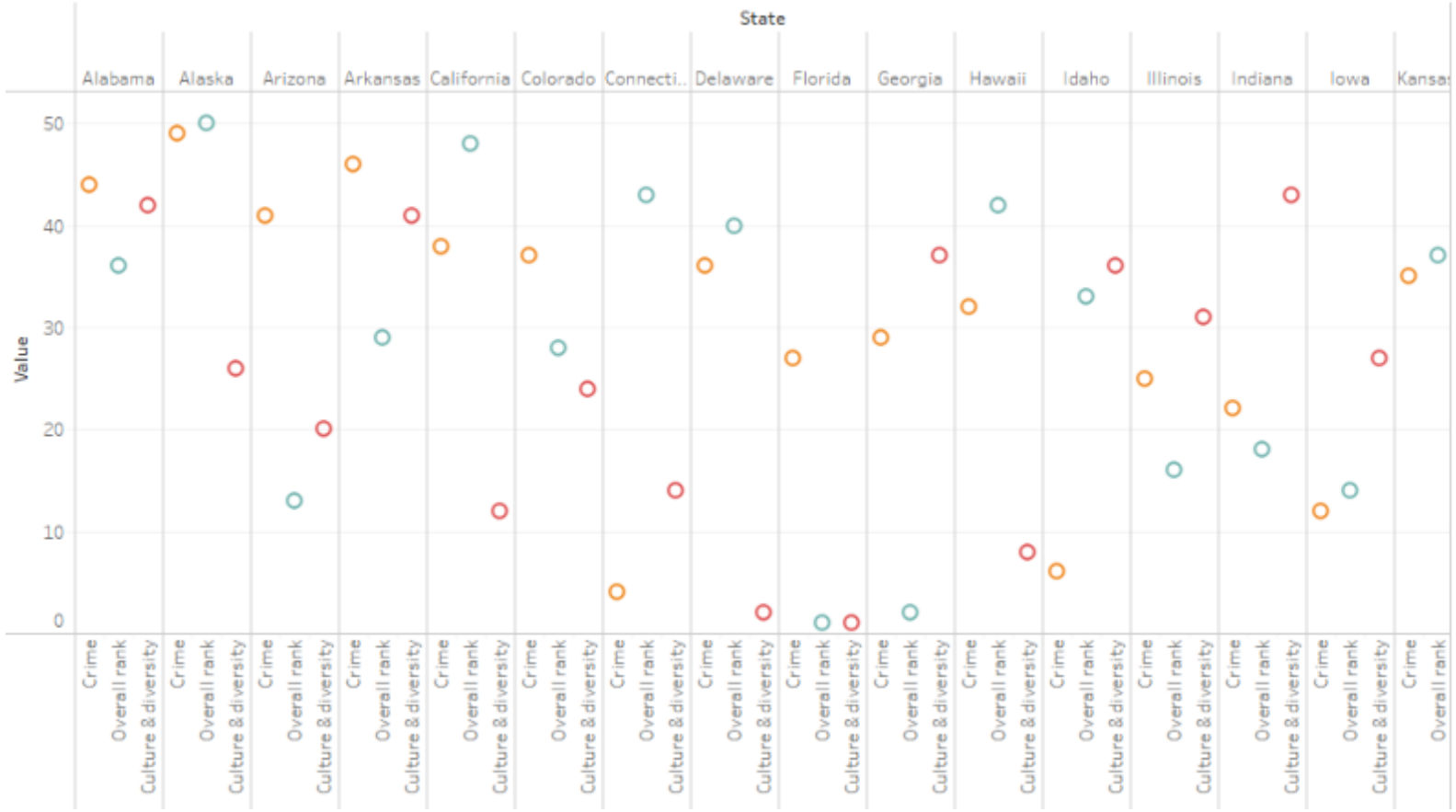
Side-by-side Bar Chart



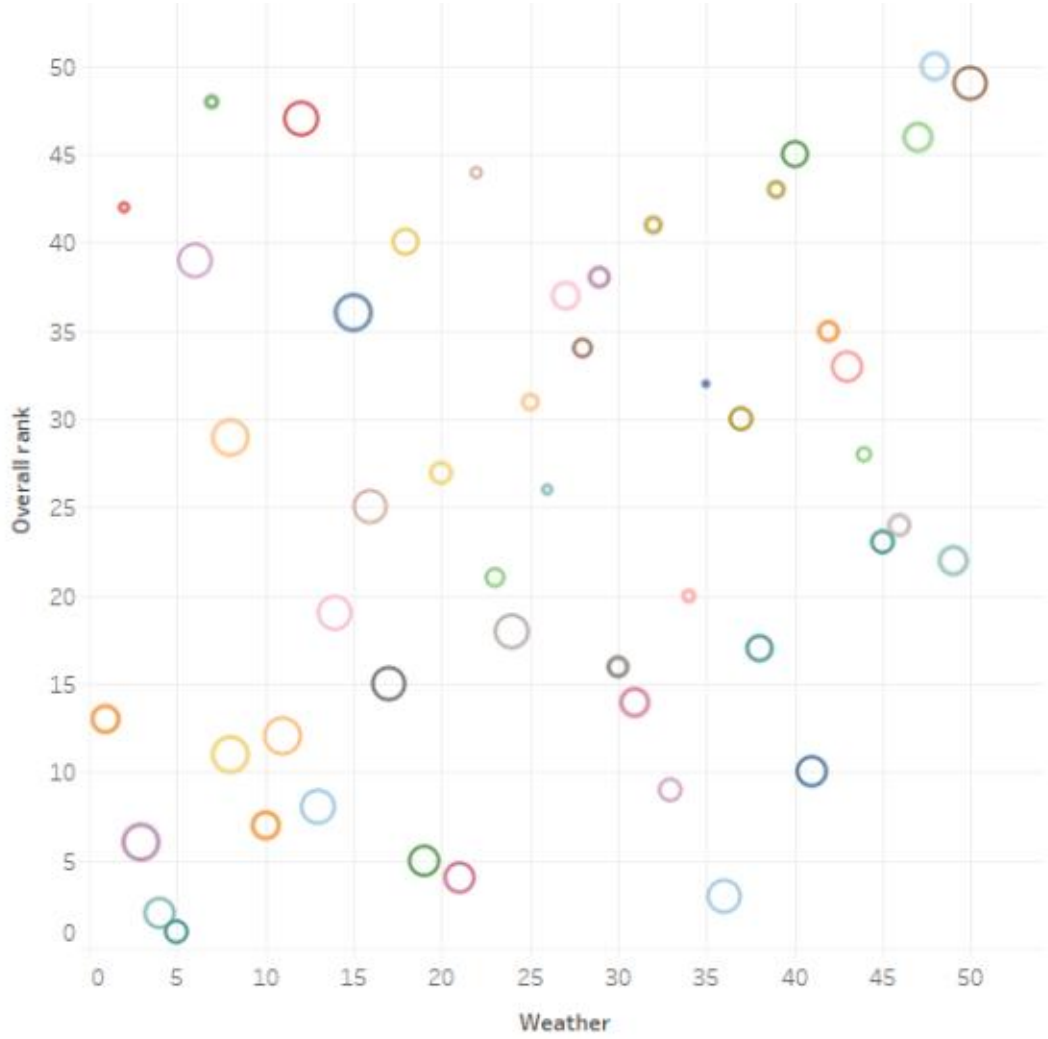
Tree Map


Alaska
50
Vermont
45
Delaware
40
Rhode Island
34
Idaho
33
Utah
31
Nebraska
30
Maine
49
Maryland
44
Louisiana
39
Arkansas
29
North Dakota
24
New
Wyoming
22
Virginia
21
California
48
Connecticut
43
Oregon
38
Colorado
28
New York
20
Illinois
16
North Carolina
15
Iowa
14
Kansas
37
Nevada
27
Oklahoma
19
West
New Mexico
47
Hawaii
42
Alabama
36
New Jersey
26
Indiana
18
Texas
7
Montana
46
Washington
41
Minnesota
35
South Carolina
25
Wisconsin
17

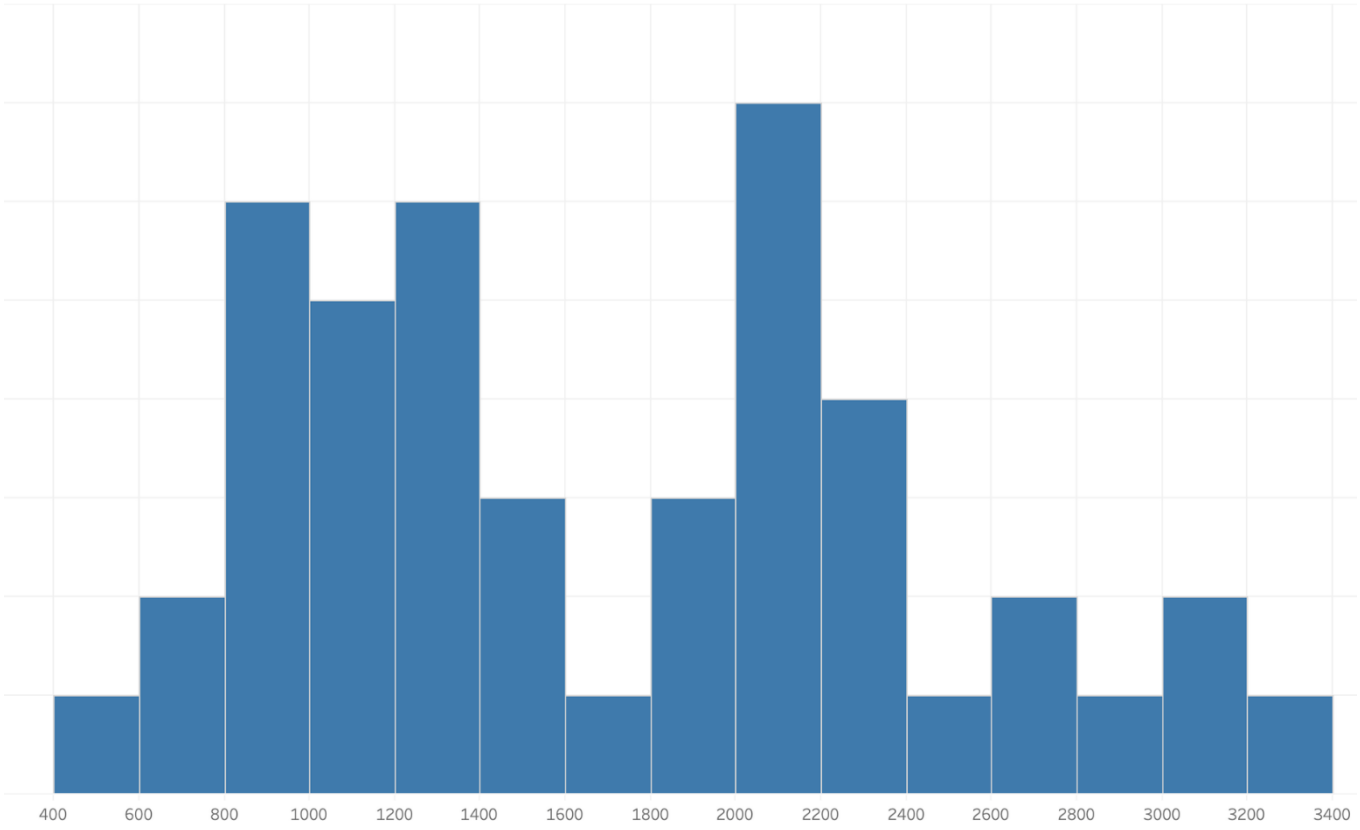
Side-by-side Circle View



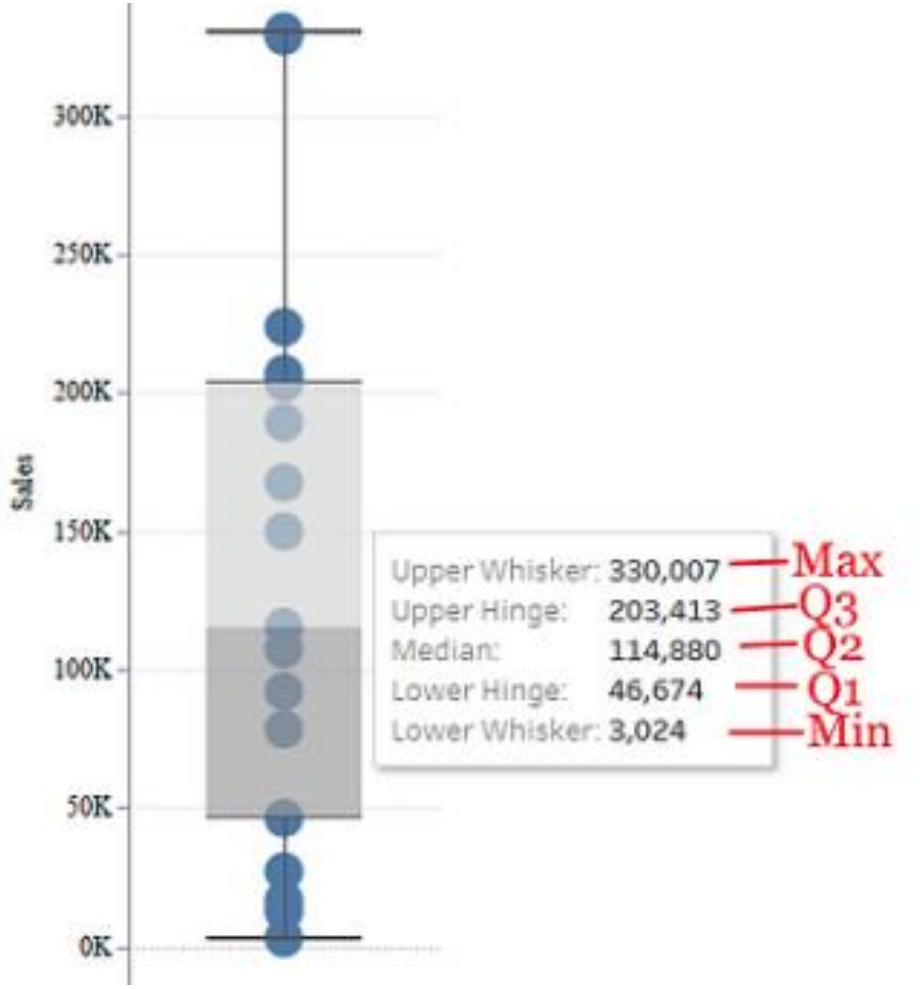
Scatter Plot



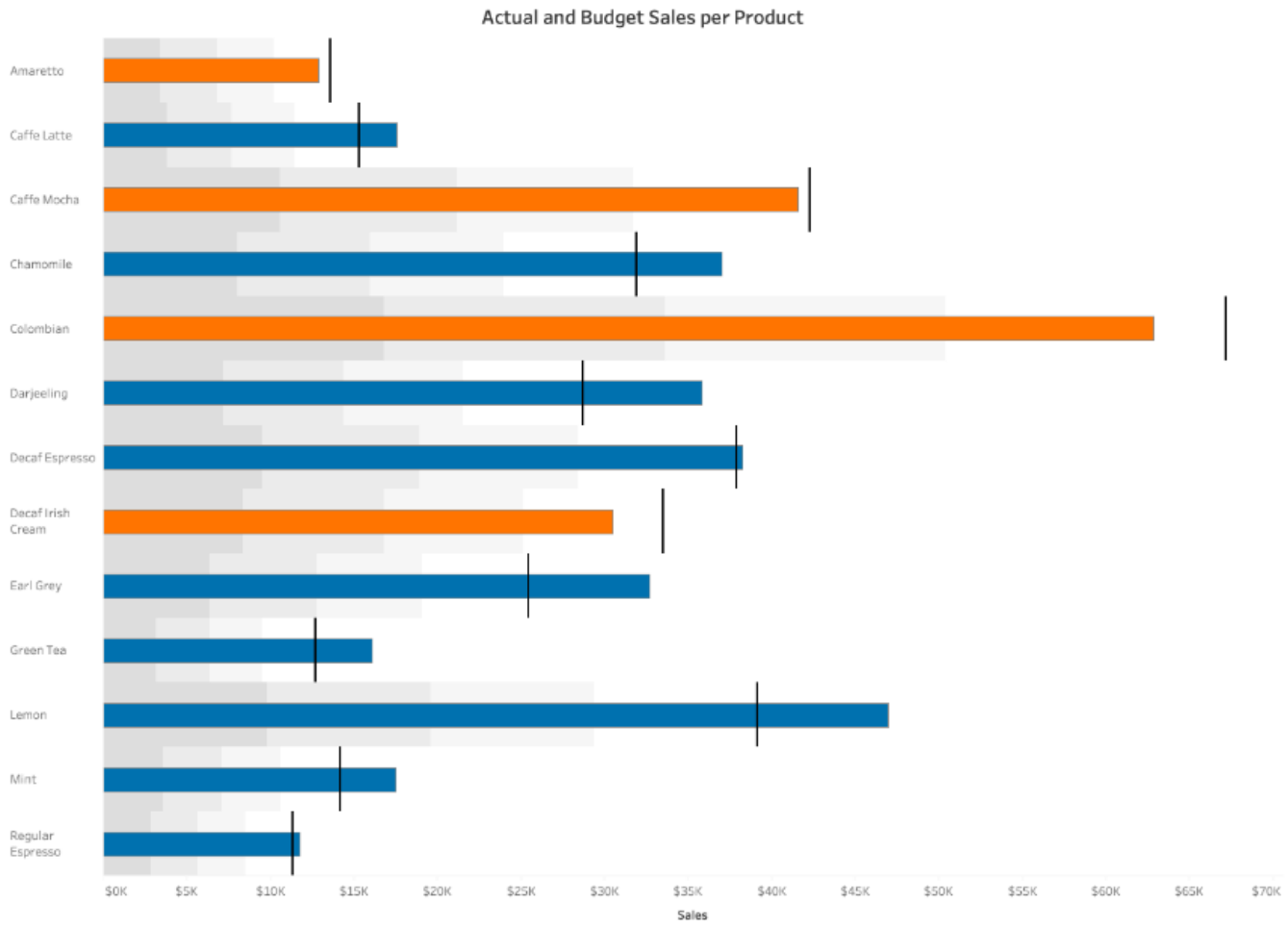
Histogram


400
600
800
1000
1200
1400
1600
1800
2000
2200
2400
2600
2800
3000
3200
3400

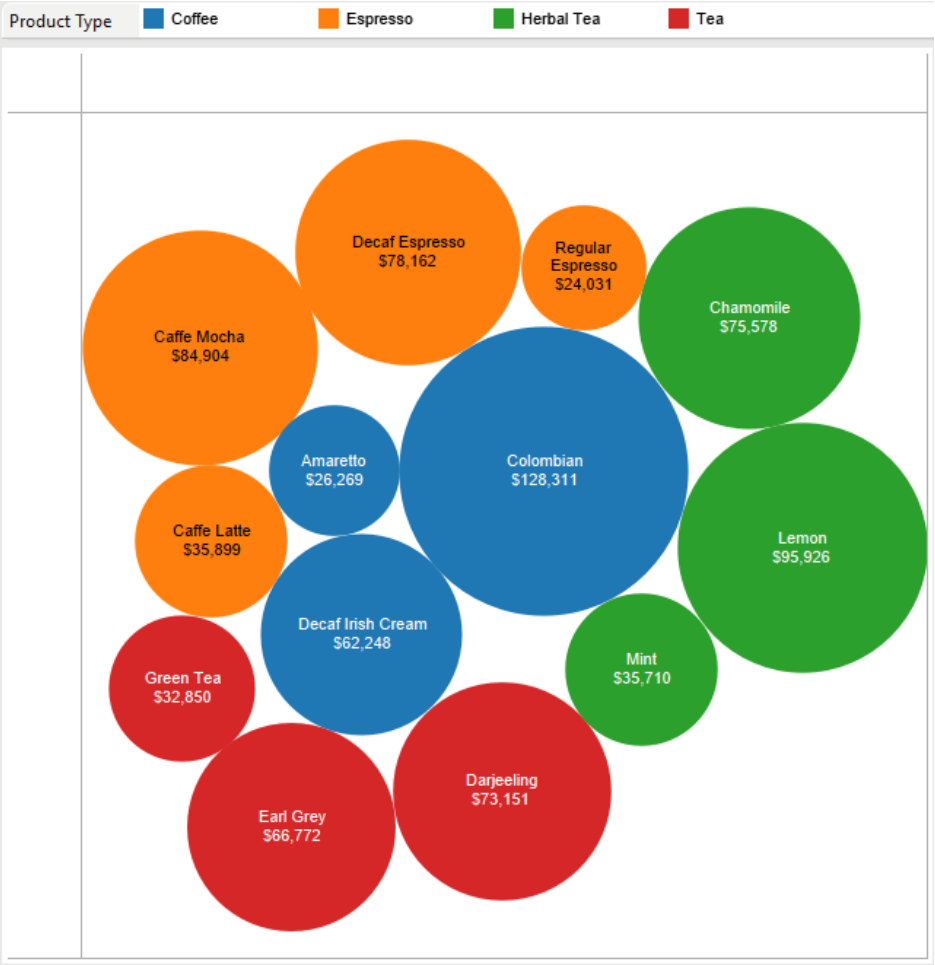
Box-and-whisker Plot



Bullet Graph



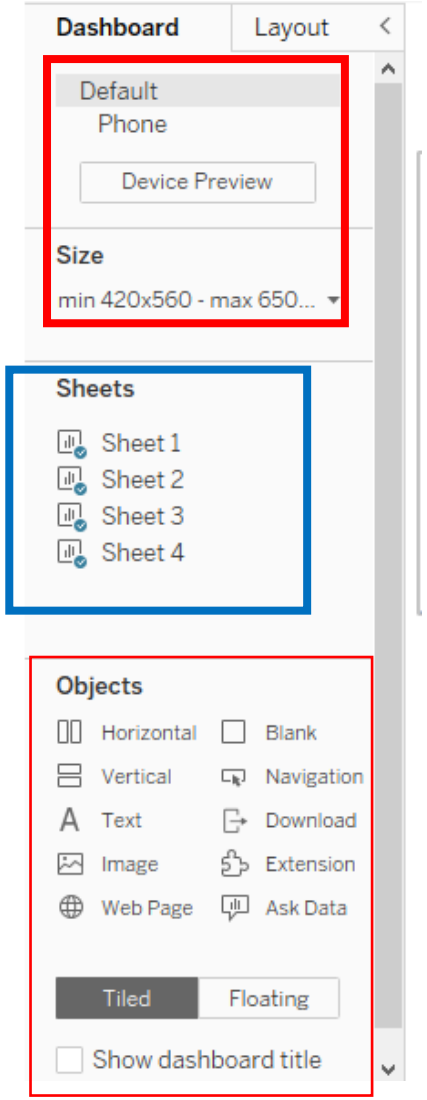
Packed Bubble Chart

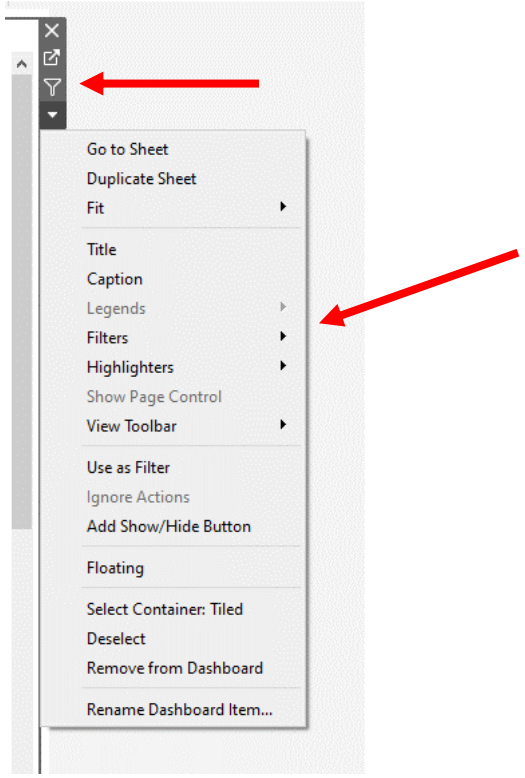


Other charts

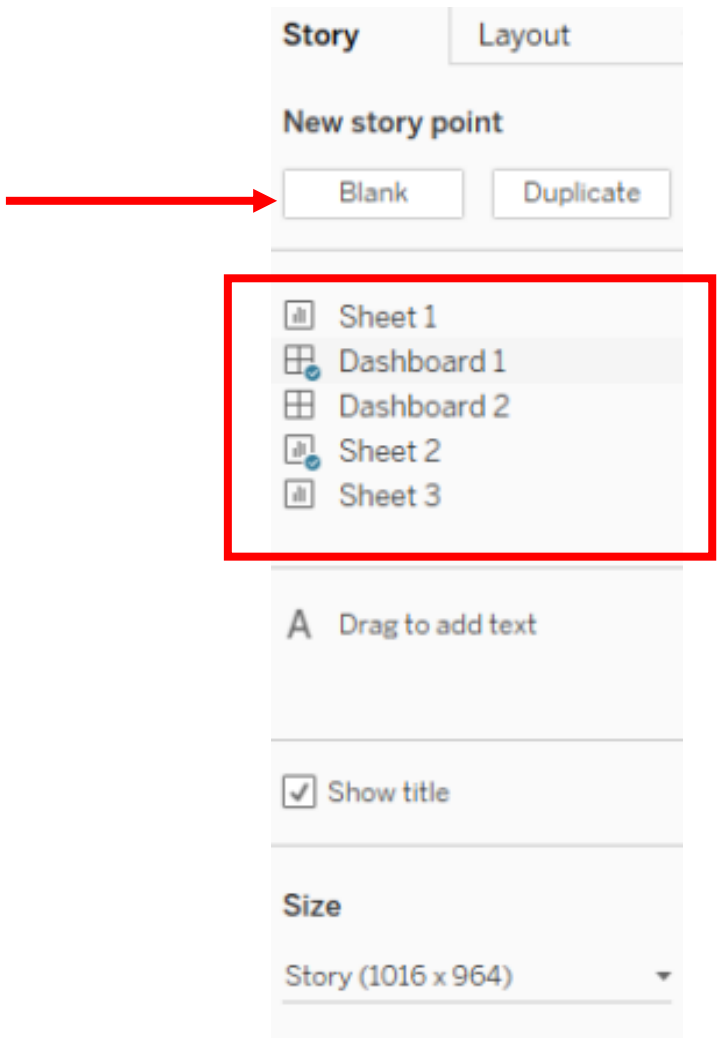
- Charts requiring a time/date dimension
  - Line charts
  - Area charts
  - Dual combination charts

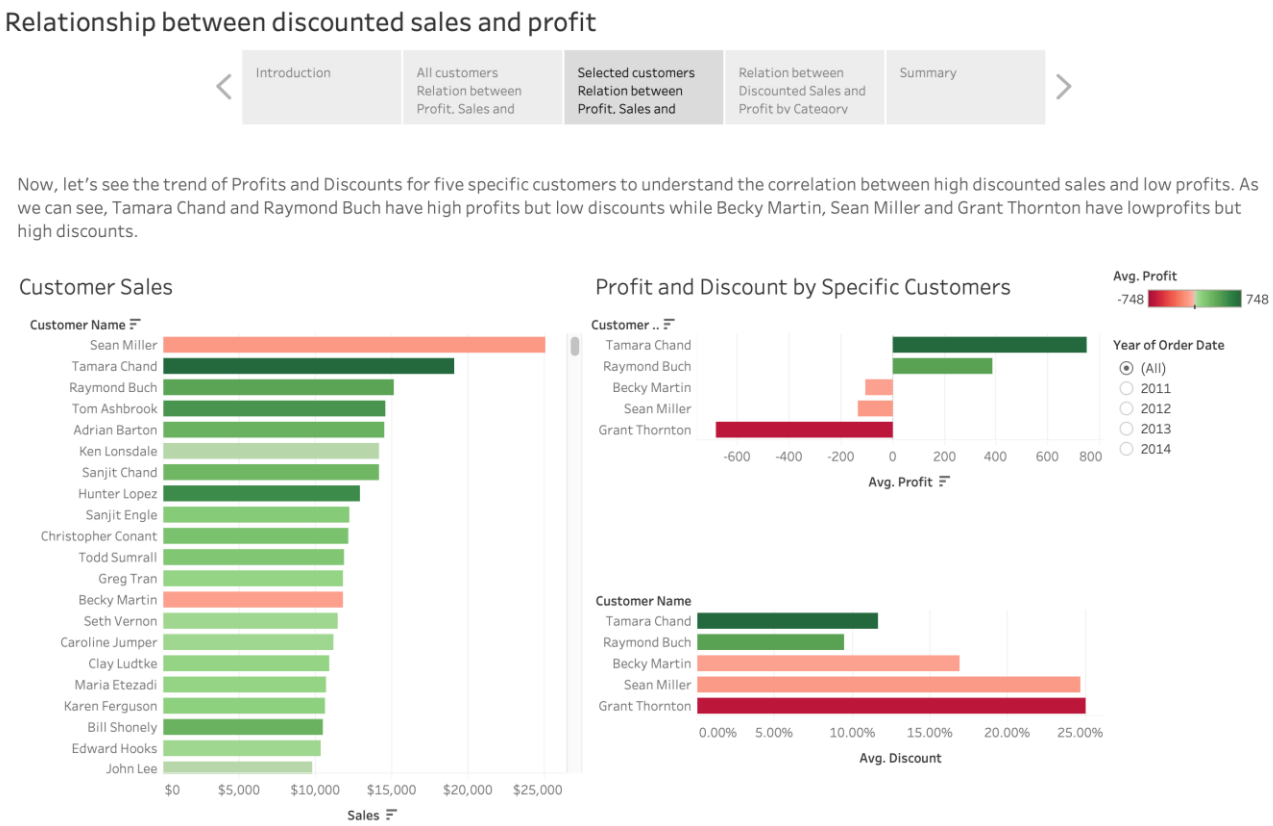
Dashboard


Dashboard
Layout
Default
Phone
Device Preview
Size
min 420x560 - max 650...
Sheets
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Objects
Horizontal
Blank
Vertical
Navigation
Text
Download
Image
Extension
Web Page
Ask Data
Tiled
Floating
Show dashboard title


Go to Sheet
Duplicate Sheet
Fit
Title
Caption
Legends
Filters
Highlighters
Show Page Control
View Toolbar
Use as Filter
Ignore Actions
Add Show/Hide Button
Floating
Select Container: Tiled
Deselect
Remove from Dashboard
Rename Dashboard Item...

Story


Story
Layout
New story point
Blank
Duplicate
Sheet 1
Dashboard 1
Dashboard 2
Sheet 2
Sheet 3
Drag to add text
Show title
Size
Story (1016 x 964)


Relationship between discounted sales and profit
Introduction
All customers Relation between Profit, Sales and
Selected customers Relation between Profit, Sales and
Relation between Discounted Sales and Profit by Category
Summary
Now, let's see the trend of Profits and Discounts for five specific customers to understand the correlation between high discounted sales and low profits. As we can see, Tamara Chand and Raymond Buch have high profits but low discounts while Becky Martin, Sean Miller and Grant Thornton have lowprofits but high discounts.
Customer Sales
Customer Name
Sean Miller
Tamara Chand
Raymond Buch
Tom Ashbrook
Adrian Barton
Ken Lonsdale
Sanjit Chand
Hunter Lopez
Sanjit Engle
Christopher Conant
Todd Sumrall
Greg Tran
Becky Martin
Seth Vernon
Caroline Jumper
Clay Ludtke
Maria Etezadi
Karen Ferguson
Bill Shonely
Edward Hooks
John Lee
$0
$5,000
$10,000
$15,000
$20,000
$25,000
Sales
Profit and Discount by Specific Customers
Avg. Profit
-748
748
Customer ..
Tamara Chand
Raymond Buch
Becky Martin
Sean Miller
Grant Thornton
-600
-400
-200
0
200
400
600
800
Avg. Profit
Year of Order Date
(All)
2011
2012
2013
2014
Customer Name
Tamara Chand
Raymond Buch
Becky Martin
Sean Miller
Grant Thornton
0.00%
5.00%
10.00%
15.00%
20.00%
25.00%
Avg. Discount